TECHNOLOGICAL UNIVERSITY DUBLIN
TALLAGHT CAMPUS

Bachelor of Science (Honours)
Bachelor of Science

Sports Science and Health

Full Time

Semester Two : May 2023

Principles of Physics

Internal Examiners
Dr Eugene Hickey

External Examiners

Day
Date May 2023
Time

Instructions to Candidates

Please answer question one and any other two questions. A table of useful equations and constants is at the end of the exam paper

Question 1 (Compulsory) (40 marks)
Answer any ten from the following sixteen parts. Each part is worth four marks

a) Express 45000J in scientific notation.
b) Convert 0.0008mm to µm
c) Convert 21000cm^3 to m^3.
d) During a Winter Olympics curling match, the stone takes 9.2s to travel the 28.5m length of the curling sheet before coming to rest. Calculate the deceleration of the stone.
e) A golf club with a clubhead of 0.35kg strikes a golf ball of mass 0.046kg. The clubhead speed before impact is 42ms^{-1}, after impact it is 33.2ms^{-1}. Calculate the launch speed of the golf ball using momentum conservation.
f) Explain the difference between static and dynamic friction.
g) What is the SI unit for work?
h) For each of the four scenarios listed below, say which type of energy is involved:
    - The energy released when we burn food during metabolism
    - The energy released from an atomic bomb
    - The energy of a car speeding on a motorway
    - The energy of a high diver at the top of a diving platform
i) A person consumes 24500 calories per day. Express this energy input in Joules.
j) Why is the Respiratory Exchange Ratio (RER) different depending on whether fats or glucose are being burnt?
k) A bicycle wheel of radius 0.2795m rotates at a frequency of 1.2 Hz. Calculate the centripetal acceleration on the rim of the wheel.
l) A ferris wheel is set in motion by an angular acceleration of 0.015rads s^{-2}. Calculate the angular velocity of the wheel at 25s and calculate how far the wheel has rotated at this time.
m) Calculate the rotational kinetic energy of a golf ball of moment of inertia 8.38x10^{-6}kgm^2 and angular velocity 335 rads/s
n) Calculate the magnitude of the vector $\vec{A} = 2\vec{x} - 4\vec{y}$.
o) Calculate the dot product, $\vec{A}.\vec{B}$, of the two vectors $\vec{A} = \vec{x} + 2\vec{y} - 5\vec{z}$ and $\vec{B} = \vec{x} - 2\vec{y} + \vec{z}$.
p) Calculate the cross product, $\vec{A}x\vec{B}$, of the two vectors $\vec{A} = \vec{x} - \vec{y} - \vec{z}$ and $\vec{B} = \vec{x} + 2\vec{y} + \vec{z}$.

Question 2 **(30 marks)**

a) Complete the following unit conversions:
   i) Express 8.5×10^{-5}kg in mg (2 marks)
   ii) Express 2.1×10^{7}nm in cm (2 marks)
   iii) Express 1.12×10^{-6}m^2 in mm^2 (2 marks)
   iv) Express 9.04g/cm^3 in kg/m^3 (2 marks)

b) Mention three types of forces and for each give an example of when it can be seen in action (2 marks)

c) What are the SI units for force? (2 marks)

d) The equation for the force of friction between two horizontal surfaces is: $F_{fric} = \mu mg$. Explain the different terms in this equation (5 marks)

e) What is the range of values for the coefficient of friction? (3 marks)

f) Friction causes a hockey puck to glide to rest across ice with a deceleration of 0.15m/s². If the puck has a mass of 0.53kg, show that the size of the force of friction on the puck is 0.0795N. (5 marks)

g) Use the results from part (f) to calculate the coefficient of friction for steel on ice. (5 marks)

Question 3 **(30 marks)**

a) What is meant by the *Moment of Inertia* of a body? (3 marks)

b) Why is the moment of inertia of a solid sphere less than that of a hollow sphere of the same mass and radius? (3 marks)

c) In soccer, a corner kick is being taken. A force of 65N is exerted at the edge of the football which has a radius of 0.110m and a mass of 0.43kg. The force is exerted for a time of 0.13s.
   i) Show that the acceleration of the ball as it is being kicked is 151 ms^{-2}. (3 marks)
   ii) Show that the velocity of the ball just after it has been kicked is 19.65ms^{-1}. (3 marks)
   iii) Calculate the kinetic energy of the football just after it has been kicked. (3 marks)
   iv) Show that the torque on the football is 7.15Nm. (3 marks)
   v) Show that the ball's moment of inertia is 3.47×10^{-3}kgm^2. (3 marks)
   vi) Show that the angular acceleration of the football as it is being kicked is 2.06×10^{3} rads s^{-2}. (3 marks)
   vii) Show that the angular velocity of the ball just after it has been kicked is 268 rads s^{-1}. (3 marks)
   viii) Calculate the rotational kinetic energy of the football just after it has been kicked. (3 marks)

Question 4 **(30 marks)**

a) Which of the following operations on two vectors produce an output vector?

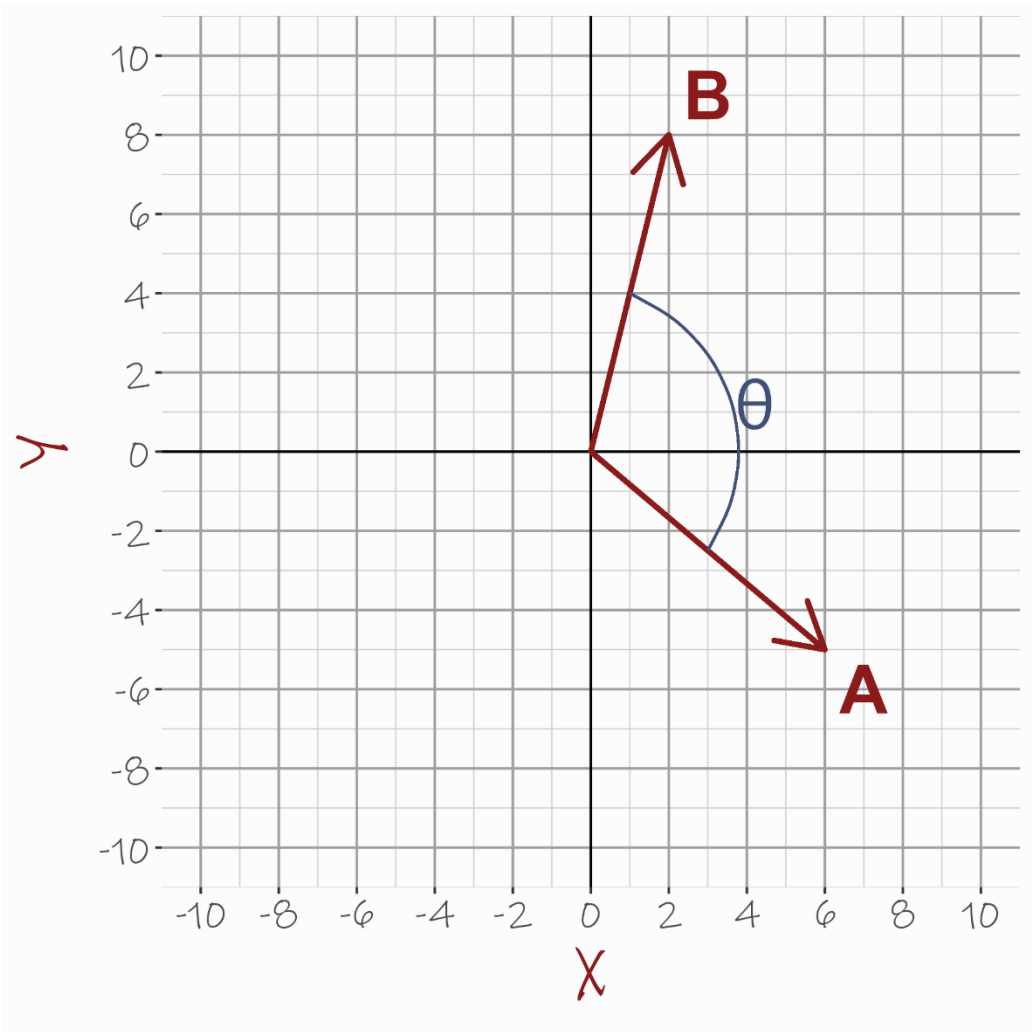
i) the dot product of two vectors, $\vec{A}\bullet\vec{B}$.

ii) the cross product of two vectors, $\vec{A}\times\vec{B}$.

iii) the magnitude of a vector, $|\vec{A}|$. (3 marks)

b) The angular momentum, $\vec{L}$, of an object moving with momentum, $\vec{p}$, on a circle of radius, $\vec{r}$, is given by , $\vec{L} = \vec{r}\times\vec{p}$. Draw a diagram showing $\vec{L}$, $\vec{r}$, and $\vec{p}$ using the right-hand rule to give their correct relative directions. (6 marks)

c) For the two vectors shown below:



i) Calculate their components, A_x, A_y, B_x, and B_y. (3 marks)

ii) Calculate their dot product, $\vec{A}\bullet\vec{B}$. (3 marks)

iii) Calculate the magnitude of $\vec{A}$ (3 marks)

iv) Calculate the magnitude of $\vec{B}$. (3 marks)

v) Calculate the cosine of the angle between $\vec{A}$ and $\vec{B}$. (3 marks)

vi) Calculate the angle, θ, between $\vec{A}$ and $\vec{B}$. (3 marks)

vii) Calculate the size and direction of the cross product, $\vec{A}\times\vec{B}$ (3 marks)

Useful Equations & Constants

Density = mass / volume

$v = v_0 + at$

$s = v_0 t + \frac{1}{2} at^2$

$v^2 = v_0^2 + 2as$

$s = (v + v_0)t/2$

$p = mv$

$W = mg$

$F_{fric} = \mu F_{reac} = \mu mg$

$P.E. = mgh$

$K.E. = \frac{1}{2} mv^2$

$$COR = \frac{v_{A_{final}} - v_{B_{final}}}{v_{A_{initial}} - v_{B_{initial}}}$$

$$COR = \sqrt{\frac{bounce\ height}{drop\ height}}$$

$\omega = 2\pi f$

$T = 1/f$

$v = \omega r$

$a = v\omega = \omega^2 r = v^2/r$

$T = F d$

$\omega = \omega_0 + \alpha t$

$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$

$\omega^2 = \omega_0^2 + 2\alpha\theta$

$\theta = (\omega + \omega_0)t/2$

$L = I \omega$

$T = I\alpha$

$I_{disc} = \frac{1}{2}MR^2$

$I_{bat} = \frac{1}{3}MR^2$

$I_{hollow\text{-}ball} = \frac{2}{3}MR^2$

$I_{solid\text{-}ball} = 2/5\ MR^2$

$RKE = \frac{1}{2}I\omega^2$

$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y$

$\vec{A} \cdot \vec{B} = |\vec{A}|\,|\vec{B}| \cos\theta$

$$\vec{A} \times \vec{B} = (A_y B_z - A_z B_y)\vec{x} + (A_z B_x - A_x B_z)\vec{y} + (A_x B_y - A_y B_x)\vec{z}$$

$g = 9.81 m/s^2$

1 calorie = 4184J

Energy production: 21.4kJ / L of O_2